THE SHEAF WITH EXACT SUPPORT OF A PSEUDO-COHERENT CLOSED IMMERSION COMMUTES WITH THE DERIVED TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E2N (sources/stacks-duality/pinned-original-source.tex line 2642).

Allowed standard prerequisites (editorial).

- The adjunction and the computation for the sheaf with exact support. Same chapter, Lemmas labelled lemma-sheaf-with-exact-support-adjoint and lemma-sheaf-with-exact-support-ext, cited at proof lines 2675-2680. Convert the isomorphism into a map given by the counit.
- Quasi-coherence of the sheaf with exact support and the algebraic tensor evaluation. Same chapter, Lemma labelled lemma-sheaf-with-exact-support-quasi-coherent, cited at proof lines 2686-2688, and More on Algebra, Lemma labelled more-algebra-lemma-internal-hom-evaluate-tensor-isomorphism part (3), cited at proof lines 2688-2690. Give the isomorphism after the affine translation.
- The stalk level alternative. Same chapter, Lemma labelled lemma-flat-bc-sheaf-with-exact-support, cited at proof lines 2690-2692. Provides the second route.

Source reference. Original lemma 0A76: sources/stacks-duality/pinned-original-source.tex, source lines 2325–2325. Same-chapter pointer into the author environment lemma at source line 2325; invoked at proof line 2660 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A77: sources/stacks-duality/pinned-original-source.tex, source lines 2339–2339. Same-chapter pointer into the author environment lemma at source line 2339; invoked at proof line 2661 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A78: sources/stacks-duality/pinned-original-source.tex, source lines 2373–2373. Same-chapter pointer into the author environment lemma at source line 2373; invoked at proof line 2665 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATK: sources/stacks-chow/tags.txt, source lines 13618–13618. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2667 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E2M: sources/stacks-duality/pinned-original-source.tex, source lines 2595–2595. Same-chapter pointer into the author environment lemma at source line 2595; invoked at proof line 2670 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $i : Z \to X$ be a pseudo-coherent closed immersion of schemes. Let $M \in D_{QCoh}(\mathcal{O}_X)$ locally have tor-amplitude in $[a, \infty)$. Let $K \in D^+_{QCoh}(\mathcal{O}_X)$. Then there is a canonical isomorphism*

$$R\mathcal{H}om(\mathcal{O}_Z, K) \otimes^{\mathbf{L}}_{\mathcal{O}_Z} Li^*M = R\mathcal{H}om(\mathcal{O}_Z, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} M)$$

in $D(\mathcal{O}_Z)$.

Proof. A map from LHS to RHS is the same thing as a map

$$Ri_*R\mathcal{H}om(\mathcal{O}_Z, K) \otimes^{\mathbf{L}}_{\mathcal{O}_X} M \longrightarrow K \otimes^{\mathbf{L}}_{\mathcal{O}_X} M$$

by Lemmas 0A76 and 0A77. For this map we take the counit $Ri_*R\mathcal{H}om(\mathcal{O}_Z, K) \to K$ tensored with id_M. To see this map is an isomorphism under the hypotheses given, translate into algebra using Lemma 0A78 and then for example use More on Algebra, Lemma 0ATK part (3). Instead of using Lemma 0A78 you can look at stalks as in the second proof of Lemma 0E2M. □